Interpretable Machine Learning for Rolling Element Bearing Fault Detection via Tree Based Models and Explainability Analysis

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***Abstract*— This research presents an advanced framework for bearing diagnostic transparency using model-agnostic explainability analysis. We conduct a systematic investigation comparing Random Forest (RF) and XGBoost classifiers on the Case Western Reserve University (CWRU) rotating machinery dataset comprising 2,300 vibration samples distributed across 10 distinct bearing conditions including normal operation and nine fault modalities. RF achieved 95.8% classification accuracy with ROC-AUC of 0.718 and PR-AUC of 0.631, while XGBoost attained 97.1% accuracy with ROC-AUC of 0.759 and PR-AUC of 0.671. Comprehensive explainability analysis employing SHAP (SHapley Additive exPlanations) values on 200 test samples reveals fundamental divergence in feature prioritization: Random Forest emphasizes interaction effects and extreme values, whereas XGBoost leverages distribution moments and central tendency measures. Critical findings indicate that inner race degradation is predominantly characterized by mean amplitude shifts, outer race damage manifests through elevated kurtosis signatures, and ball defects exhibit combined indicator dependencies. These interpretable insights provide actionable guidance for maintenance personnel, demonstrating that transparent decision-making can be successfully reconciled with strong predictive capability in industrial maintenance contexts.**

Keywords— bearing fault detection, explainable AI, machine learning, random forest, gradient boosting

I. INTRODUCTION

Rolling-element bearing failures represent a critical challenge in modern industrial systems, contributing to approximately 40-50% of unexpected rotating machinery interruptions and resulting in substantial economic losses [1] . Beyond the immediate component replacement costs, bearing faults cascade throughout production environments, causing extended downtime periods, supply chain disruptions, and creating safety hazards for personnel working in proximity to faulty equipment [2]. Conventional maintenance philosophies, which is reactive repair strategies that address failures after occurrence or time-based preventive replacement schedules that replace components arbitrarily, prove increasingly inadequate for contemporary industrial requirements where production efficiency and operational continuity demand precision-based decision making.

Predictive maintenance strategies represent a fundamental paradigm shift away from purely reactive and preventive approaches, enabling condition-based decision-making informed by real-time equipment assessment and degradation monitoring [3] . Recent machine learning advances have demonstrated remarkable potential for bearing condition classification, with deep learning architectures achieving accuracy rates exceeding 99% on benchmark datasets through convolutional neural network architectures and transfer learning methodologies [4]. Despite these impressive numerical results, significant adoption barriers persist within real- world manufacturing environments. Industrial stakeholders remain skeptical of high-accuracy black-box systems offering predictions without transparent reasoning or explainable decision pathways, creating organizational resistance to autonomous diagnostic deployment and limiting practical real-world implementation [5].

This investigation directly addresses the critical explainability gap by implementing comprehensive SHAP-based transparency on inherently interpretable machine learning models. Rather than pursuing marginal accuracy improvements at the cost of interpretability loss, we prioritize transparent decision-making mechanisms that maintenance teams can understand, validate, and trust in operational contexts [6]. We systematically examine Random Forest and XGBoost classifiers, comparing their diagnostic mechanisms and feature utilization patterns through comprehensive explainability analysis.

The principal contribution of this work involves demonstrating how different algorithmic architectures construct divergent decision pathways while maintaining competitive predictive performance on bearing diagnostics. Detailed SHAP analysis reveals fault-type- specific feature patterns precisely aligned with underlying bearing physics, enabling development of maintenance strategies adapted to specific degradation mechanisms and severity progression patterns.

II. RELATED WORK

A. Machine Learning for Bearing Fault Detection

Contemporary bearing fault detection literature emphasizes deep learning methods targeting accuracy optimization. Convolutional neural networks have emerged as dominant architectures, with Guo et al. demonstrating hierarchical adaptive deep CNN (ADCNN) achieving 97.7%

accuracy through hierarchical layer adaptation to fault characteristics [7]. Hybrid architectures combining CNN with LSTM networks exploit complementary strengths—CNNs capture local spatial features while LSTM captures temporal dependencies—yielding 99.6% accuracy through parallel feature extraction branches [8]. Transfer learning via fine-tuned ResNet-50 converts 1D vibration signals to 2D spectrograms enabling ImageNet pretrained feature exploitation, achieving 99.99% accuracy [9]. Beyond pure accuracy, ensemble methods combining multiple base learners have demonstrated robustness benefits. A hybrid LSTM-random forest model enhanced by grey wolf optimization demonstrated effective multiple bearing fault detection through ensemble combination of deep sequential models with tree-based classifiers [10].

B. Explainability and Interpretability

Interpretability remains critical for industrial implementation despite ensemble method popularity. SHAP (SHapley Additive exPlanations) provides model-agnostic explanations grounded in cooperative game theory's Shapley values, quantifying individual feature contributions to predictions [11]. SHAP advantages include: (1) theoretical foundation in Shapley values ensuring fair and robust explanations; (2) model-agnostic applicability across linear, tree-based, and neural network architectures; (3) unified framework generating both global explanations (aggregate feature importance) and local explanations (instance-specific reasoning) [12]. Research demonstrates SHAP-ranked features enabling bearing fault classification exceeding 90% accuracy with substantially reduced feature sets (5-10 features versus 100+ original features), facilitating edge device deployment [13].

XGBoost combined with SHAP explanations has shown effectiveness across diverse domains. Chen et al. developed XGBoost-based breast cancer diagnosis achieving 98.42% accuracy with SHAP explanations identifying highest-impact features [14]. Machine learning-driven approaches for arterial functional aging assessment via XGBoost achieved AUC of 0.8722 with identified top-five predictive features [15].

III. METHODOLOGY

A. Dataset Characterization

This work utilizes the CWRU benchmark dataset containing 2,300 balanced vibration recordings distributed uniformly across 10 bearing condition categories [16]. The dataset encompasses one normal or healthy operational state and nine distinct fault modalities representing systematic combinations of three spatial locations (inner race, outer race, rolling element) and three defect magnitudes (0.007, 0.014, 0.021 inch fault diameters).

Vibration data acquisition employed a 2 HP Reliance Electric motor operating at four distinct load configurations corresponding to shaft speeds of approximately 1797, 1772, 1750, and 1730 RPM respectively. Acceleration sensors positioned at both drive-end and fan-end bearing locations captured high-resolution data at 48 kHz sampling frequency, yielding signal segments of 2,048 points representing 0.042 seconds of continuous vibration acquisition. This temporal window provides sufficient frequency resolution for capturing bearing characteristic frequencies within the critical 500 Hz to 10 kHz operational band where bearing fault signatures concentrate.

B. Feature Characterization

Nine fundamental time-domain statistical attributes were systematically extracted from each vibration segment: peak value, minimum point, mean amplitude, standard deviation, crest factor, root mean square (RMS), form factor, skewness coefficient, and kurtosis moment [17]. This comprehensive feature set captures amplitude behavior patterns, variability characteristics, distribution shape properties, and impulsiveness indicators critical for accurate bearing fault diagnosis. Time-domain feature extraction successfully circumvents computational overhead associated with frequency transformation techniques, enabling straightforward deployment on edge computing platforms and IoT devices with constrained computational resources [18], making this methodology particularly suited for Industry 4.0 environments.

C. Data Preprocessing and Stratification

The 2,300-sample dataset underwent carefully planned stratified 67.4-32.6 (training-test) partitioning, deliberately preserving fault- class proportions throughout both subsets to ensure unbiased performance evaluation. Standard scaling normalization transformed each feature using exclusively training-set statistics. Test set normalization was performed using only training parameters to rigorously prevent data leakage contamination [19].

D. Model Development and Hyperparameter Optimization

Random Forest Classifier was configured with 100 decision trees, maximum depth of 20, and minimum samples per leaf of 2. Grid search optimization systematically evaluated depth values {10, 15, 20, 25} and minimum sample counts {1, 2, 4} through 10-fold stratified cross-validation performed exclusively on training data.

XGBoost Classifier employed 150 sequential boosting rounds with learning rate 0.05, maximum tree depth 6, minimum child weight 1, and L2 regularization parameter . Hyperparameter tuning employed Bayesian optimization over learning rates {0.01, 0.05, 0.1, 0.2}, maximum depths {3, 6, 9, 12}, and regularization parameters {0.1, 1, 10}. Both models incorporated probability calibration through softmax output normalization enabling multi-class classification across 10 bearing condition categories [20].

E. SHAP Explainability Framework

SHAP values were computed using TreeExplainer algorithms, leveraging efficient tree traversal computation optimized specifically for tree-based ensemble models. Analysis focused on 200-sample test subsets randomly selected from the full 750-sample test set while maintaining proportional fault-class representation. For each sample instance, SHAP values quantified individual feature contributions toward each 10 class predictions, generating three-dimensional explanation matrices [20].

IV. RESULTS & ANALYSIS

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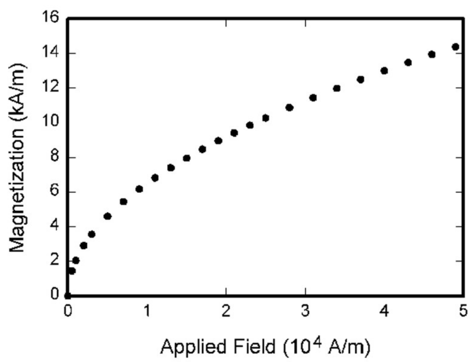

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